Episode_2

Finding Biomedical Papers: PubMed & PMC

> ⚠ **Warning**
>
> This page never shows real results or paper details. When you click *Open in PubMed*, the live database runs the search, so your result counts will differ from anyone else's and change over time. Expected results below describe *what kind* of papers to look for, not specific titles or numbers.

1. Introduction

	PubMed	PubMed Central (PMC)
What it is	An index of biomedical *citations and abstracts* (title, authors, journal, abstract).	A free archive of *full-text* articles.
Searches	Titles, abstracts, keywords, MeSH indexing terms.	The whole article text (methods, figures, tables, references).
Access	Free to search; the paper itself may sit behind a journal paywall.	Free to read.
Best for	Surveying a topic, finding reviews, seeing what exists.	Finding a method, a mutation, a figure, or a phrase buried in the body of a paper.
Coverage	Broad.	Only articles deposited in PMC (many journals, funder-mandated papers, preprints for some programmes).

Rule of thumb: start in PubMed to map the field. Move to PMC when you need the full text or very specific details that abstracts omit. Every PMC article also has a PubMed record, linked by a PMID (PubMed ID) and a PMCID (e.g. “PMC” followed by digits).

2. Basic Search Methods

Step 2.1: A simple keyword search

1. Open pubmed.ncbi.nlm.nih.gov.
2. Type words into the big box and press **Search**.
3. PubMed automatically maps your words to indexing terms and also searches all fields. You can see how it interpreted your text under **Search details** in the right-hand sidebar (you may need to scroll).

Step 2.2: Gene names and synonyms

The same gene or protein is written in several ways. OXA-48 appears as “OXA-48”, “OXA 48”, “blaOXA-48”, or “OXA-48-like”. TP53 appears as “TP53”, “p53”, “tumor protein p53”, and “Trp53” (mouse). If you search only one, you miss papers. Join synonyms with `OR` (see Section 3). Look up official names and aliases in NCBI Gene (TP53 record).

Step 2.3: Refine with the left-hand tools

- **Results by year** bar chart: click a range to narrow dates.
- **Filters** (left): Free full text, Article type (Review, Clinical Trial…), Publication date, Species. Click *Additional filters* for more.
- **Sort by**: Best match, Most recent, Publication date.

3. Advanced Search Methods

Step 3.1: Advanced Search Builder

1. Go to pubmed.ncbi.nlm.nih.gov/advanced (or click *Advanced* under the search box).
2. In **Builder**, pick a field (e.g. *Title/Abstract*), type a term, and click **Add** or **Add with AND / OR / NOT** for each additional row.
3. Click **Search**, or **Add to history** to save without running.

Step 3.2: Boolean operators (must be CAPITALS)

Operator	Effect	Example
`AND`	Both ideas required; narrows	`OXA-48 AND Klebsiella`
`OR`	Either idea; widens (use for synonyms)	`TP53 OR p53`
`NOT`	Excludes; use sparingly, it can remove relevant papers	`TP53 NOT mice`

Use **parentheses** to group: `(TP53 OR p53) AND (breast OR mammary)`.

Step 3.3: Phrase searching

Put exact phrases in double quotes: `"molecular dynamics"`. Without quotes the words can appear anywhere. Note: if a quoted phrase is not in PubMed's phrase index, PubMed may warn you. Try a close variant.

Step 3.4: Field tags

Put a tag in square brackets right after a term to restrict where it is searched.

Tag	Meaning	Example
`[tiab]`	Title or abstract	`OXA-48[tiab]`
`[ti]`	Title only	`TP53[ti]`
`[mh]`	MeSH term (controlled vocabulary)	`"Tumor Suppressor Protein p53"[mh]`
`[pt]`	Publication type	`review[pt]`
`[dp]`	Publication date	`2020:2026[dp]`
`[au]`	Author	`Smith J[au]`
`[ta]`	Journal	`Antimicrob Agents Chemother[ta]`

Browse MeSH terms in the MeSH Database. Note: tags like `[tiab]` switch off PubMed's automatic term mapping, so include your own synonyms.

Step 3.5: Filters

Use the left sidebar after searching. Tick *Free full text* to see only freely readable papers (many are in PMC). Filters stay applied until you click *Reset all filters*.

Step 3.6: Search History

1. On the Advanced page, scroll to **History and Search Details**. Every search gets a number (#1, #2…).
2. Combine earlier searches: type `#1 AND #2` into the Builder box and search.
3. Click ⋯ beside a line for details or to delete it. History is kept for about 8 hours of inactivity unless you are signed in to a free My NCBI account (optional).

Try it: run `OXA-48[tiab]` (#1), run `review[pt]` (#2), then search `#1 AND #2`.

4. Practical Examples: OXA-48

Each example has a goal, an exact query, steps, what to expect, and how to adjust the size of your results.

5. Practical Examples: TP53

6. PubMed vs. PMC in Practice

Part A: From a PubMed record to full text

1. Search PubMed with this query, then pick any result:
2. Look for a green **Free PMC article** label under the title in the results list. Click the title to open the record.
3. On the record page, the right-hand **Full text links** box lists options: *Free PMC article*, publisher site, etc. Click *Free PMC article* to open the full text in PMC.
4. If there is no PMC link, the paper may only be at the publisher. Check the publisher link for “Open Access” or “Free” labels, or try the same title in PMC (Part B).
5. Back on the PubMed results page, tick the **Free full text** filter to hide everything you cannot read.

Part B: Searching PMC directly

1. Open pmc.ncbi.nlm.nih.gov.
2. Paste a query and search. PMC searches the *entire text*, so common words give huge, noisy results. Quote phrases and combine specific terms:
3. To narrow to where a term is the main subject, add a field such as `[Title]` or `[Abstract]`. Example:
4. Open a result and use the **Figures**, **Tables** and **References** sections in the page outline. Click **PubMed** or the PMID link on the article page to jump back to its PubMed record.

Interpreting PMC results: a hit means your words appear *somewhere* in the paper, including the reference list or a side comment. Check the abstract and headings to see whether the paper is truly about your topic. If you have too many results, add the `[Title]` or `[Abstract]` restriction. If too few, remove a restriction or add synonyms with `OR`.

7. Cheat Sheet & Troubleshooting

Problem	Try
Too many results	Add `AND` terms; use `[tiab]`; add a date or article-type filter.
Too few results	Add synonyms with `OR`; remove quotes or field tags; drop a restrictive filter.

Problem	Try
Irrelevant papers	Restrict to `[ti]`; use `NOT` carefully; read *Search details* to see how PubMed interpreted the query.
Can't open the paper	Use *Free full text* filter or look for *Free PMC article*.
Zero results	Check spelling, quote marks, and bracket tags. Capitalise `AND/OR/NOT`.

Official help: PubMed User Guide · PMC User Guide. Interfaces and filter names are occasionally updated by NCBI, so labels may differ slightly from this page.